Geotopia University

Professor’s Note

"Imagination is more important than knowledge. For knowledge is limited, whereas imagination embraces the entire world, stimulating progress, giving birth to evolution."

- Albert Einstein

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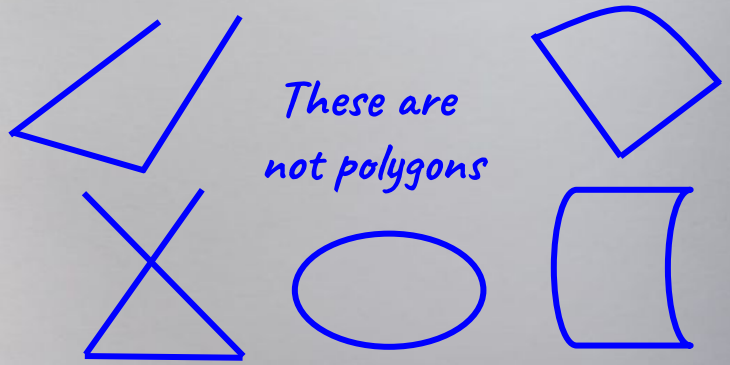
Note #3 Polygons

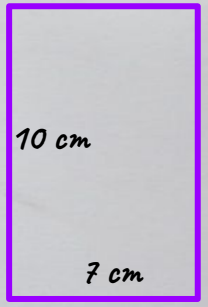
These are polygons



These are not polygons

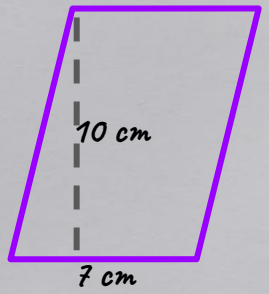
Area of Rectangle:
Length times width

$10 \times 7 = 70\ cm^2$



Area of Parallelogram:
Base times height

$10 \times 7 = 70\ cm^2$



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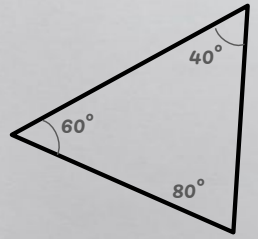
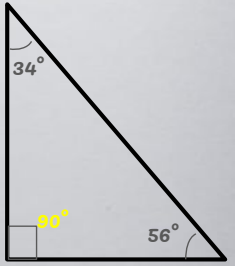
Note #8 Triangles

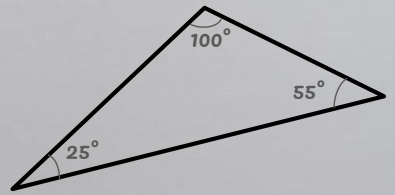
Acute Triangle



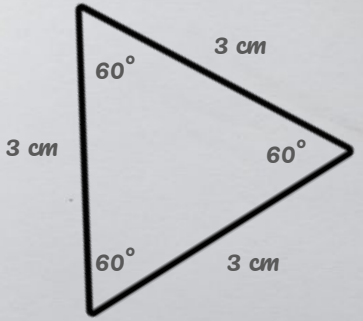
Right Triangle



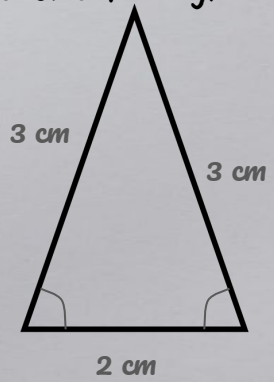
Obtuse Triangle



Equilateral Triangle



Isosceles Triangle



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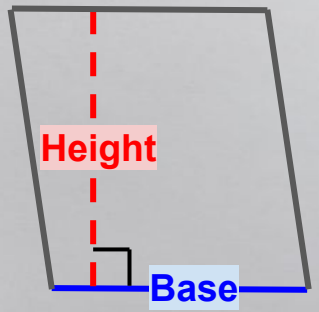
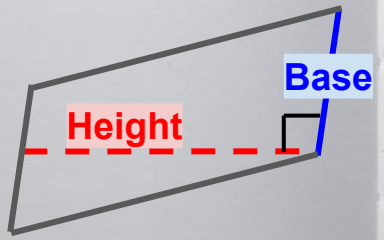
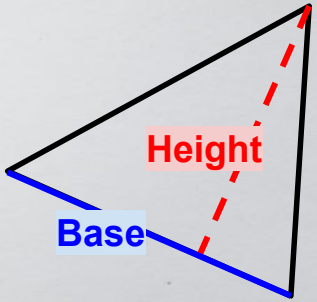
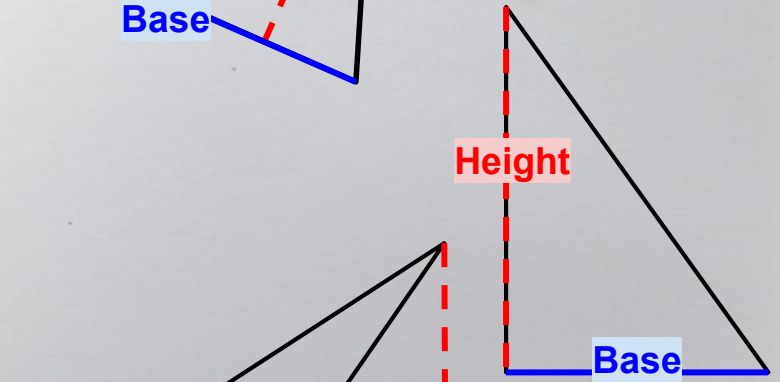
Note #13 Base and Height

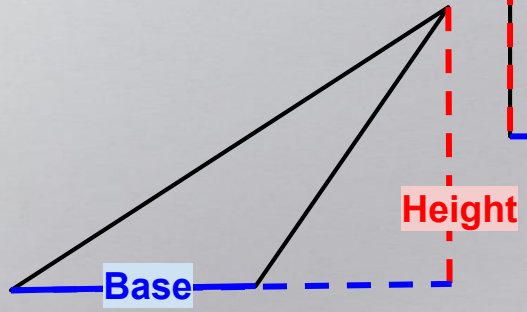
Base and height always form a 90° angle! (right angle)











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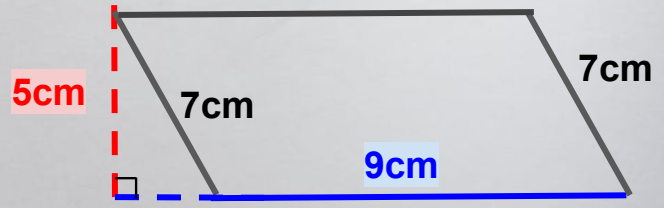
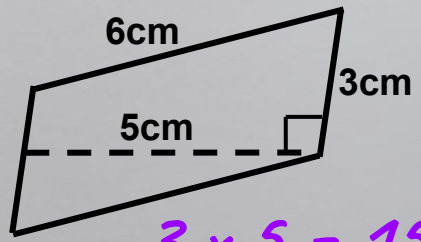
Note #15 Area

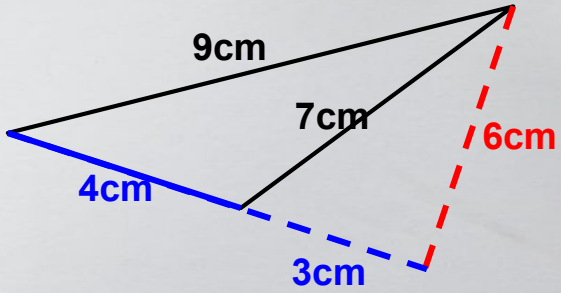


Area of Parallelogram = Base times height

$9 \times 5 = 45\ cm^2$

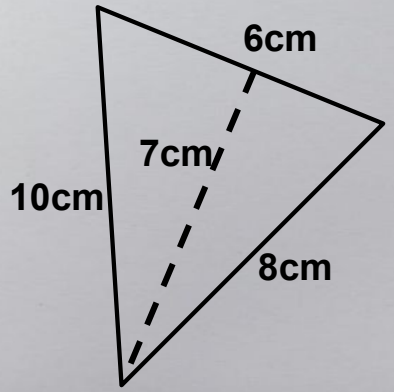


$3 \times 5 = 15\ cm^2$



Area of triangle = half of (Base times height)

$4 \times 6 \div 2 = 12\ cm^2$



$6 \times 7 \div 2$

$= 21\ cm^2$

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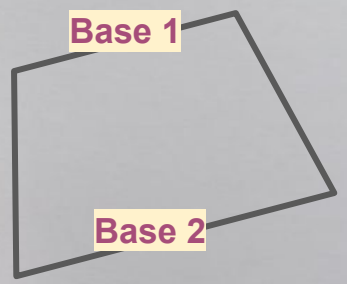
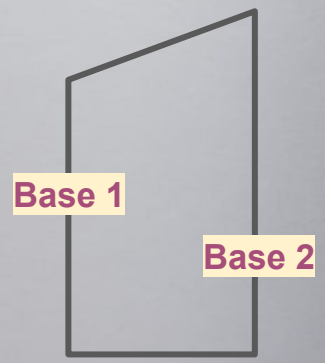
Note #20 Area of trapezoid

There are 2 bases in a trapezoid.
They are base 1 and base 2.
They are parallel to each other.

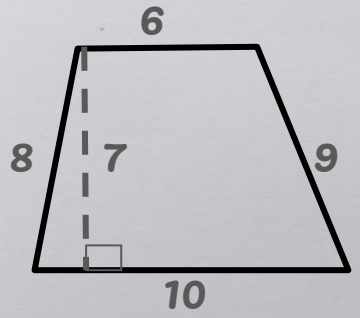
(Parallel lines are going to the same direction and never go across each other)





Area of trapezoid =

(Base 1 + Base 2) x Height ÷ 2



$(6 + 10) \times 7 \div 2 = 56\ cm^2$

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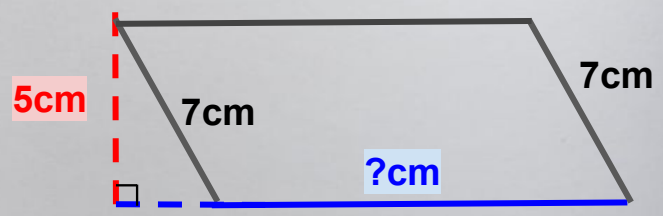
Note #15 Finding missing length

The area is 45 cm^2

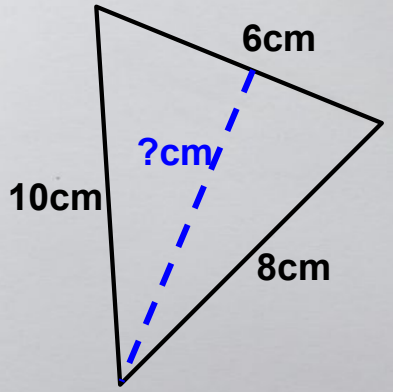


The missing base is

$45 \div 5 = 9$ cm

Area Height Base

The area is 12 cm^2



The missing height is

$12 \times 2 \div 6 = 4$ cm

Double the area Base Height

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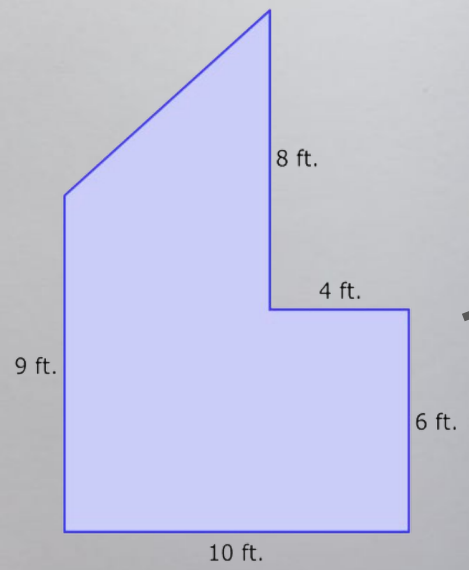
Note #23 Composite Figure

How can we find the area of this composite figure?



Cut!

Triangle A:

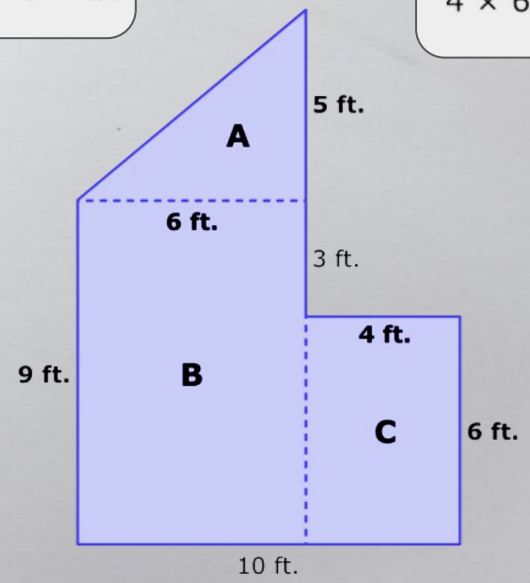
$\frac{1}{2} \times 6 \times 5 = 15$

Rectangle B:

$9 \times 6 = 54$

Rectangle C:

$4 \times 6 = 24$



$15 + 54 + 24 = 93\ ft^2$

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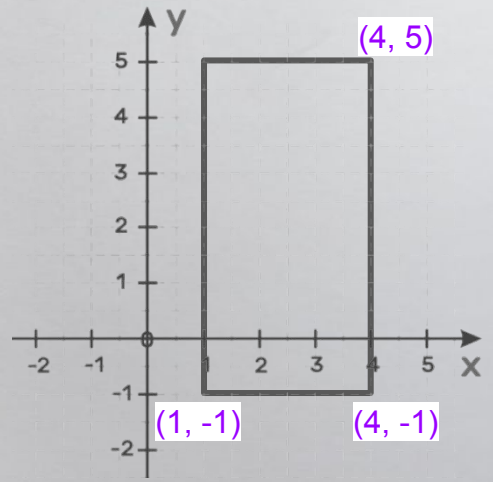
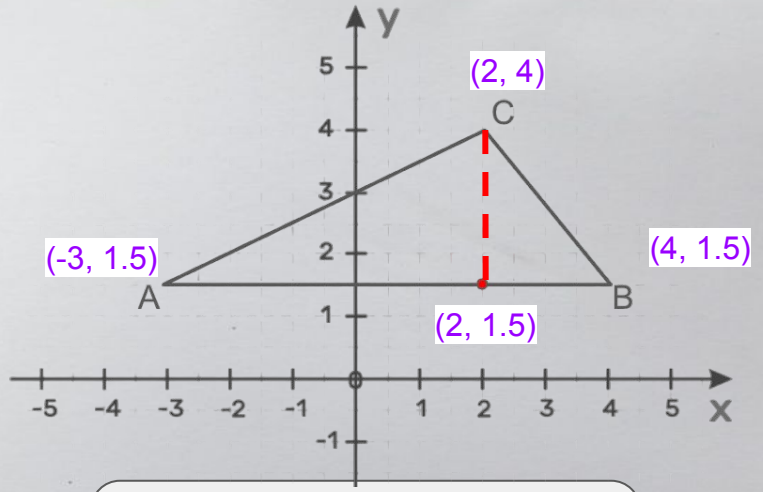
Note #29 Coordinate Plane



From (1, -1) to (4, -1)
The base is 3

From (4, 5) to (4, -1)
The base is 6

Area= 6 x 3 = 18



From (-3, 1.5) to (4, 1.5)
The base is 7

From (2, 4) to (2, 1.5)
The height is 3.5

Area= 3.5 x 7 ÷ 2 = 12.25

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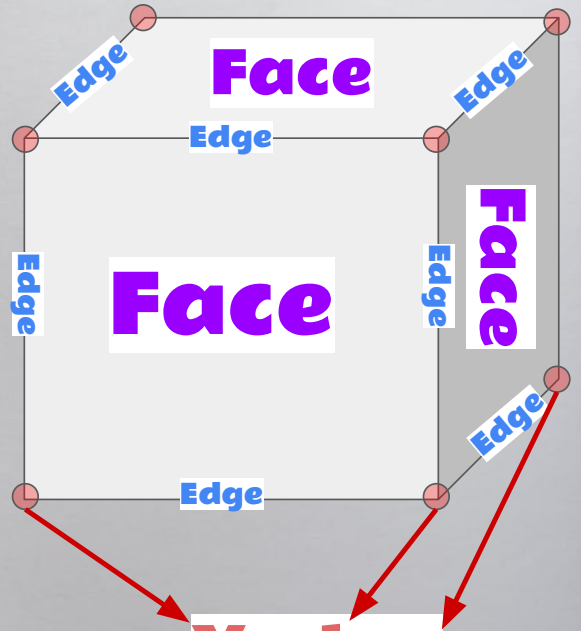
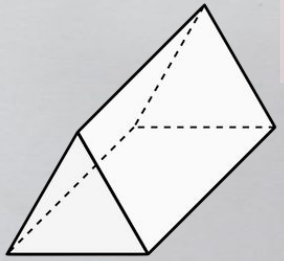
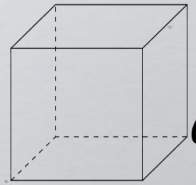
Note #35 3D Shapes



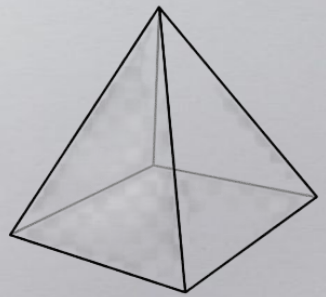
Rectangular Prism



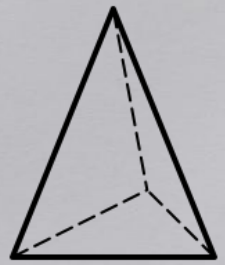
Triangular Prism



Cube(Square Prism)



Square Pyramid



Triangular Pyramid

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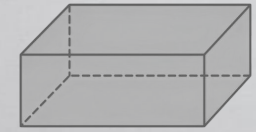
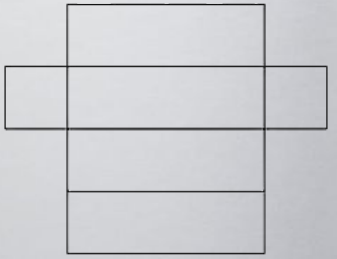
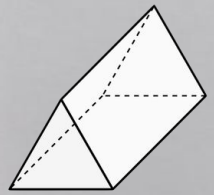
Note #36 Nets



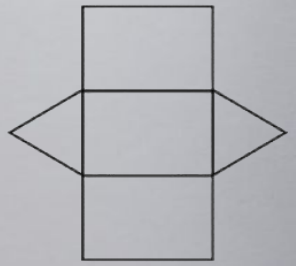
Rectangular Prism



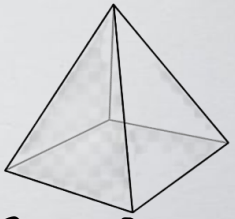
Nets



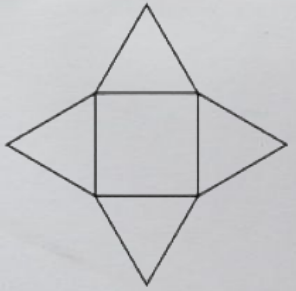
Triangular Prism



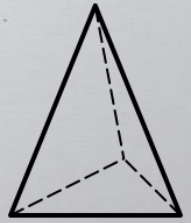
Nets



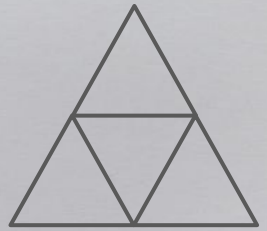
Square Pyramid



Nets



Triangular Pyramid



Nets

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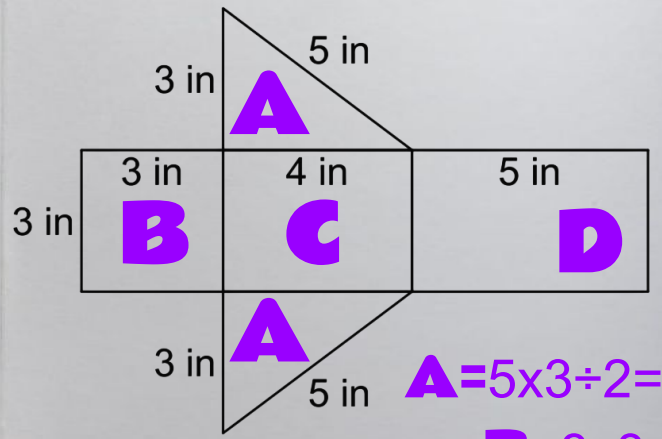
Note #46 Surface Area

To find the surface area, we need to find the area of each face from its net!



A=5x3÷2=7.5

B=3x3=9

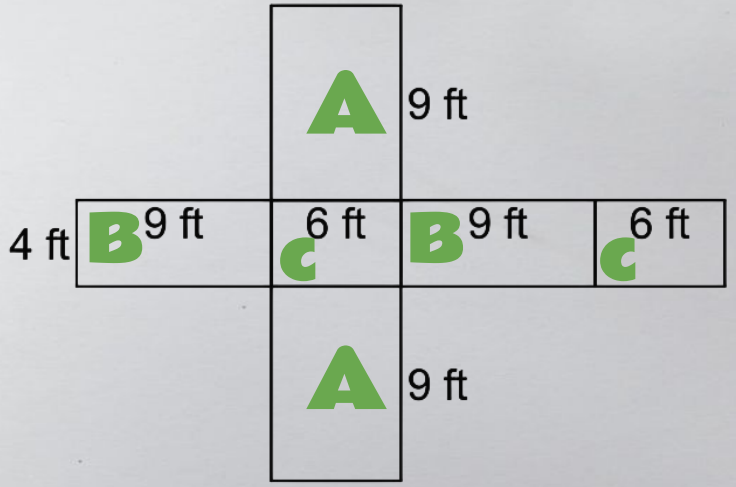
C=3x4=12

D=3x5=15

Surface Area=

7.5+7.5+9+12+15= 51 in²

(two triangles)



A=9x6=54

B=9x4=36

C=6x4=24

Surface Area=

54+54+36+36+24+24= 228 ft²

(two same faces)

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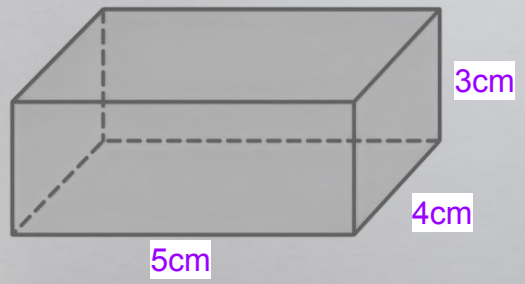
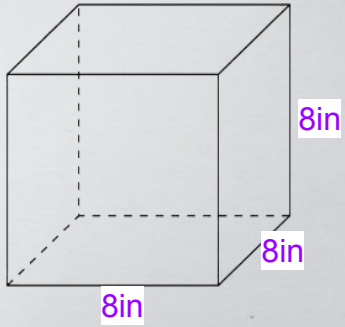
Note #57 Volume

To find the volume, we need to find the base area and then **multiply** by the height!

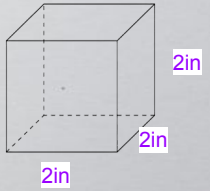


Volume= (5 x 4) x 3 = 60 cm^3

Base area Height

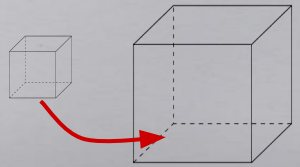


Volume=
(8 x 8) x 8
= 512 cm^3



Volume=
(2 x 2) x 2 = 8 cm^3

How many small cubes (2-inch side length) can fit in the big cube (8-inch side length)?



512÷8
=64 cubes!

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